

Cyclistic Bike-Share Case Study

How Do Annual Members and Casual Riders Use Cyclistic Bikes Differently?

Prepared for: Lily Moreno, Director of Marketing, and the Cyclistic Executive Team

1. Business Task

This report analyzes how annual members and casual riders use Cyclistic bikes differently, in order to help the marketing team, led by Director Lily Moreno, design a strategy to convert casual riders into annual members. Cyclistic's finance team has established that annual members are significantly more profitable than casual riders, making this conversion a strategic priority for the company's future growth.

Key stakeholders: Lily Moreno (Director of Marketing) and the Cyclistic Executive Team, who must approve any resulting marketing recommendations.

2. Data Sources

This analysis uses twelve months of Cyclistic's historical trip data (July 2025 - June 2026), made publicly available by Motivate International Inc. under an open data license.

- Original & first-party: collected directly by the bike-share system via GPS and docking-station logs.
- Comprehensive: covers a full 12-month cycle across all ride types and both user categories.
- Current: the most recent 12 months of available data at the time of analysis.
- Cited: sourced and licensed transparently, as documented by the provider.
- Reliable: presumed reliable given automated data collection, though this could not be independently verified.

Key limitation: the data contains no personally identifiable information (PII). Each row represents a ride record, not a unique rider — it is not possible to determine how many individual people are behind these figures, or whether a single rider took multiple trips.

3. Data Cleaning Summary

Trip data was processed and cleaned using Microsoft Excel (for an initial subset of months) and Python (pandas) for the remaining months, following an identical, validated cleaning methodology across all twelve files.

- Two calculated fields were added to every ride: ride length (duration) and day of week.
- Rides with negative duration (a data error) were checked for; none were found.
- Rides longer than 24 hours were excluded, as these predominantly reflect lost or stolen bikes rather than genuine usage.
- Rides shorter than 60 seconds were excluded as "false starts," consistent with the data provider's own definition.

- A small number of records (under 0.15% per month) had missing end-location coordinates; these were retained, as location data was not required for the duration/day-of-week analysis.

After cleaning, 5,764,564 valid ride records remained for analysis, with under 4% of raw records excluded as documented data-quality outliers.

4. Analysis Summary

Comparing the two rider types across four dimensions revealed a clear and consistent behavioral pattern:

Metric	Casual Riders	Annual Members
Average ride length	~19.3 minutes	~12.3 minutes
Median ride length	~11.6 minutes	~8.8 minutes
Total ride volume	2,031,799 (35%)	3,732,765 (65%)
Weekly usage peak	Saturday & Sunday	Weekdays (Tue–Wed)
Electric bike share	69.0%	65.6%
Winter usage drop	~93%	~75%

Taken together, these findings point to two distinct usage patterns. Annual members rely on Cyclistic bikes for frequent, short-duration commuting, with usage spread consistently across weekdays and throughout the year. Casual riders, by contrast, use the service in a manner consistent with recreational or leisure activity — fewer, longer rides concentrated on weekends and heavily dependent on favorable summer weather.

5. Key Visualizations

Four supporting charts accompany this report (attached separately): (1) average and median ride length by user type, (2) ride volume by day of week, (3) bike-type preference by user type, and (4) the monthly ride-volume trend across the full 12-month period.

6. Top Recommendations

1. Launch a weekend-focused membership offer, paired with an electric-bike incentive.

Casual riders concentrate their activity on weekends, with longer but less frequent rides than members, and show a slightly stronger preference for electric bikes (69.0% vs. 65.6%). We recommend designing a membership offer specifically targeted at weekend usage patterns, paired with an electric-bike-related perk to increase its appeal to this segment.

2. Time the primary marketing campaign to peak summer usage.

Casual rider activity is markedly more weather-sensitive than members' (a 93% winter drop-off, versus 75% for members). We recommend concentrating the core marketing campaign during the summer peak (July–August), when casual riders are already actively using and paying for the service — the moment they are most likely to respond to a cost-saving annual membership pitch.

3. Review operational bike availability during peak weekend/summer demand.

Casual rider demand, while smaller in total volume than members' (2.0M vs. 3.7M rides), is heavily concentrated into a narrow time window (weekends, summer months). We recommend that the operations team assess bike availability at stations near recreational destinations during these peak windows. A casual rider who cannot find an available bike doesn't just lose one ride — they lose a potential first positive experience that could otherwise support conversion to membership.